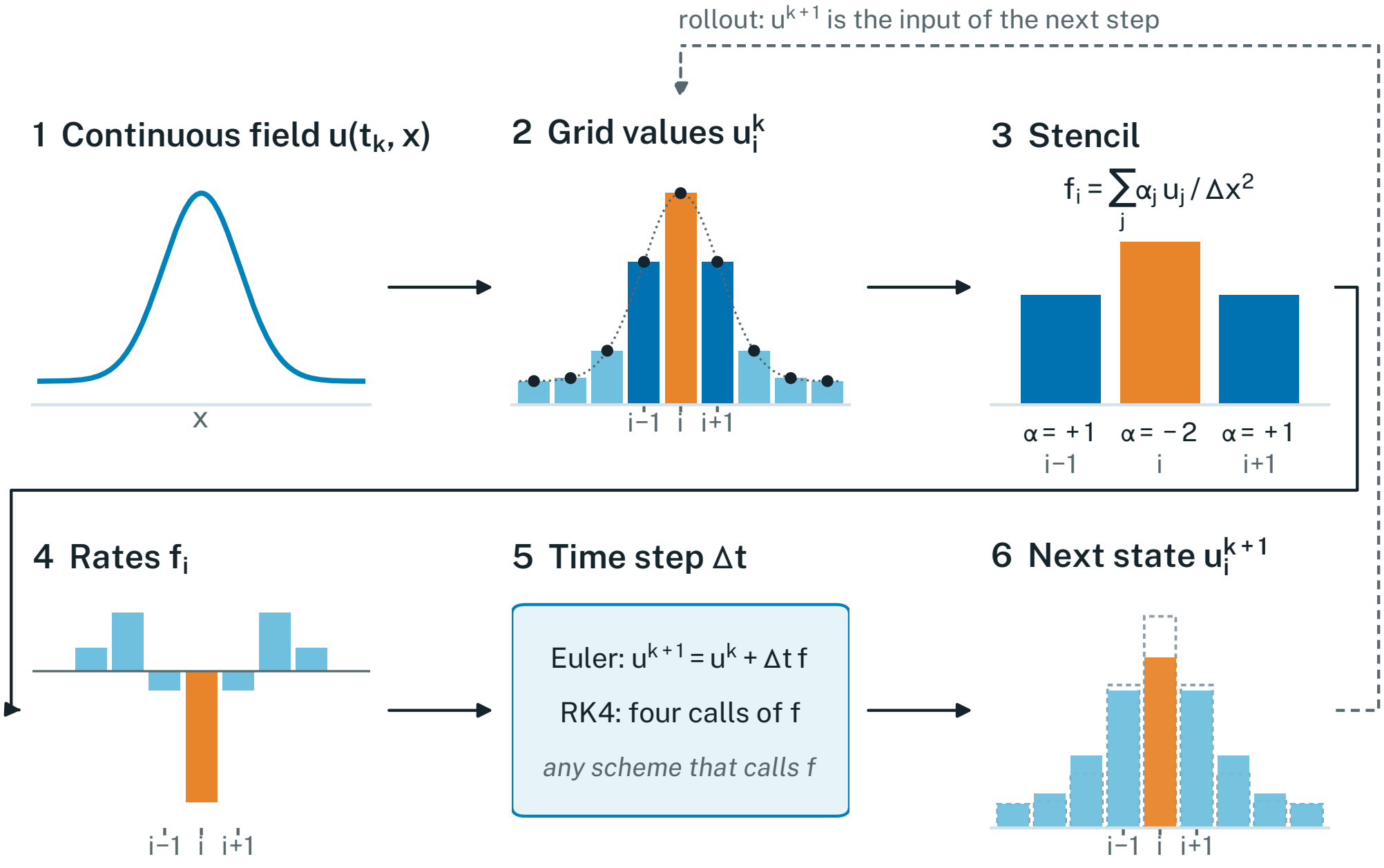
rollout: u^{k+1} is the input of the next step
1 Continuous field $u(t_k, x)$
x
2 Grid values u_i^k
i−1 i i+1
3 Stencil
$f_i = \sum_j \alpha_j u_j / \Delta x^2$
α = +1 α = −2 α = +1
i−1 i i+1
4 Rates f_i
i−1 i i+1
5 Time step Δt
Euler: $u^{k+1} = u^k + \Delta t\, f$
RK4: four calls of f
any scheme that calls f
6 Next state u_i^{k+1}
i−1 i i+1